

MH2100 Calculus III

Midterm Examination (Solutions)

AY 2025/2026, Semester 1

Quantitative Research Society @NTU

October 2025

Overview

- **Coverage:** limits and total differentiability; directional derivatives and the chain rule; extrema and Lagrange multipliers; double integrals and polar coordinates.
- **Format:** four problems, 100 marks in total; 110 minutes.
- **Source:** MH2100 Midterm Examination, AY2025/2026 Semester 1. The questions below reproduce the paper; the solutions and marking schemes are unofficial QRS@NTU material.

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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Question 1**(25 Marks)****PROBLEM 1.**

(a) Evaluate

$$\lim_{(x,y) \rightarrow (2,1)} \frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2}. \quad (5 \text{ points})$$

(b) Determine whether or not the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, defined by

$$\forall x, y \in \mathbb{R} : \quad f(x, y) \stackrel{\text{df}}{=} \sqrt[3]{x^3 + y^3},$$

is totally differentiable at $(0, 0)$.**(20 points)****Solution****Method 1: Direct computation**

(a)

$$\begin{aligned} x^2y - x^2 - 4y + 4 &= (y-1)(x-2)(x+2) \\ xy - x - 2y + 2 &= (x-2)(y-1) \\ \frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2} &= x+2 \end{aligned} \quad (x \neq 2, y \neq 1)$$

$$\boxed{\lim_{(x,y) \rightarrow (2,1)} \frac{x^2y - x^2 - 4y + 4}{xy - x - 2y + 2} = 4.}$$

(b)

$$\begin{aligned} f_x(0,0) &= \lim_{h \rightarrow 0} \frac{\sqrt[3]{h^3} - 0}{h} = 1, & f_y(0,0) &= 1 \\ \text{total differentiability} &\implies L(h,k) = h + k \\ f(t,t) &= \sqrt[3]{2}t, & L(t,t) &= 2t & ((h,k) = (t,t)) \\ \frac{f(t,t) - f(0,0) - L(t,t)}{\sqrt{t^2 + t^2}} &= \frac{(\sqrt[3]{2} - 2)t}{\sqrt{2}|t|} \not\rightarrow 0 & (t \rightarrow 0) \end{aligned}$$

$$\boxed{f \text{ is not totally differentiable at } (0,0).}$$

Marking scheme (25 marks)(a) **(5 marks)**

- 2 marks: factors the numerator correctly.
- 1 mark: factors the denominator correctly.
- 1 mark: cancels the common factors on the domain of the quotient.
- 1 mark: obtains the limit 4.

(b) **(20 marks)**

- 4 marks: computes $f_x(0,0) = f_y(0,0) = 1$.

- 4 marks: identifies the only possible derivative $L(h, k) = h + k$.
- 6 marks: chooses and evaluates a valid path, such as (t, t) .
- 4 marks: shows that the normalized remainder does not tend to zero.
- 2 marks: states the correct conclusion about total differentiability.

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Question 2**(25 Marks)****PROBLEM 2.**(a) A function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is defined by

$$\forall x, y, z \in \mathbb{R} : \quad f(x, y, z) \stackrel{\text{df}}{=} z^2 - x^3 y^2.$$

Let $\mathbf{u}$ denote the unit vector pointing from $(1, -1, 1)$ toward $(2, 1, 0)$.Compute $(D_{\mathbf{u}}f)(1, -1, 1)$ – with justification.**(10 points)**(b) Two functions, $f, g : \mathbb{R}^2 \rightarrow \mathbb{R}$, satisfy the following four properties:

- f is totally differentiable at $(1, 4)$.
- The tangent plane to the graph of f at $(1, 4, f(1, 4))$ is

$$\{(x, y, z) \in \mathbb{R}^3 \mid 3x + 5y = z + 26\}.$$

- $f_{xy}(1, 4) = f_{yx}(1, 4) = 1$ and $f_{xx}(1, 4) = f_{yy}(1, 4) = -1$.
- For all $x, y \in \mathbb{R}$, we have $g(x, y) = x \cdot f(x^2 - y, x + 2y)$.

Compute $g_{xy}(-2, 3)$ – with justification.**(15 points)****Solution****Method 1: Chain rule**

(a)

$$\mathbf{u} = \frac{(2, 1, 0) - (1, -1, 1)}{\|(1, 2, -1)\|} = \frac{1}{\sqrt{6}}(1, 2, -1)$$

$$\text{grad } f = (-3x^2 y^2, -2x^3 y, 2z)$$

 $(f \text{ differentiable})$

$$\text{grad } f(1, -1, 1) = (-3, 2, 2)$$

$$(D_{\mathbf{u}}f)(1, -1, 1) = \text{grad } f(1, -1, 1) \cdot \mathbf{u} = \frac{-3 + 4 - 2}{\sqrt{6}}$$

$$\boxed{(D_{\mathbf{u}}f)(1, -1, 1) = -\frac{1}{\sqrt{6}}.}$$

(b) $u = x^2 - y$, $v = x + 2y$ ($u_y = -1$, $v_y = 2$), $g = xf(u, v)$.

$$(x, y) = (-2, 3) \implies (u, v) = (1, 4)$$

$$z = 3x + 5y - 26 \implies f_x(1, 4) = 3, \quad f_y(1, 4) = 5$$

The data give $f_{xx}, f_{xy}, f_{yx}, f_{yy}$ at one point only. f_x, f_y need not be differentiable near $(1, 4)$, so the second-order chain rule is unjustified.Assume $f \in C^2$ near $(1, 4)$:

$$\begin{aligned} g_x &= f + x(2xf_x + f_y) \\ g_{xy} &= (-f_x + 2f_y) + x[2x(-f_{xx} + 2f_{xy}) + (-f_{yx} + 2f_{yy})] \\ &= (-3 + 10) + (-2)[(-4)(1 + 2) + (-1 - 2)] \\ &= 7 + (-2)(-15) \end{aligned}$$

$$\boxed{g_{xy}(-2, 3) = 37 \quad \text{if } f \text{ is } C^2 \text{ near } (1, 4).}$$

Counterexample: $U = s - 1$, $V = t - 4$, $F = F_0 + Q$,

$$F_0 = -3 + 3U + 5V - \frac{1}{2}U^2 + UV - \frac{1}{2}V^2, \quad Q = \frac{U^2V^2}{U^2 + V^2} \quad (Q(0, 0) = 0).$$

$Q \in C^1$, $\nabla Q(0, 0) = 0$, and the four iterated second partials of Q at $(0, 0)$ are 0. $\therefore F$ satisfies every datum.

Along $x = -2$, $y = 3 + t$: $(U, V) = (-t, 2t)$, $Q_U = -\frac{32t}{25}$, $Q_V = \frac{4t}{25}$, $Q = t^2Q(-1, 2)$.

$$Q\text{-part of } (g_x)_y(-2, 3) = \frac{d}{dt}[t^2Q(-1, 2)]_{t=0} + (-2)\left[(-4)\left(-\frac{32}{25}\right) + \frac{4}{25}\right] = 0 - \frac{264}{25}$$

$$g_{xy}(-2, 3) = 37 - \frac{264}{25} = \frac{661}{25} \neq 37$$

$\boxed{\text{underdetermined}}.$

Marking scheme (25 marks)

(a) (10 marks)

- 3 marks: finds and normalizes the direction vector.
- 4 marks: computes $\text{grad } f(1, -1, 1) = (-3, 2, 2)$.
- 2 marks: applies the directional-derivative formula with justification.
- 1 mark: obtains $-1/\sqrt{6}$.

(b) (15 marks)

- 3 marks: extracts $f_x(1, 4) = 3$ and $f_y(1, 4) = 5$ from the tangent plane.
- 2 marks: identifies $(x^2 - y, x + 2y) = (1, 4)$ at $(-2, 3)$.
- 3 marks: explains that the stated data do not justify the second-order chain rule.
- 4 marks: gives a valid counterexample or otherwise establishes that the literal question is underdetermined.
- 3 marks: under an explicit C^2 -neighbourhood assumption, differentiates g_x , applies the chain rule correctly, and obtains 37.

Question 3**(25 Marks)****PROBLEM 3.**

A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$\forall x, y \in \mathbb{R} : \quad f(x, y) \stackrel{\text{df}}{=} e^{-xy}.$$

Let $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 \leq 1\}$.

- (a) Determine all the critical point(s) of $f|_C$. **(5 points)**

Note: Every critical point of $f|_C$ belongs – by definition – to the interior of C .

- (b) Determine the nature of each critical point of $f|_C$. **(5 points)**

Note: Apply the Second-Derivative Test.

- (c) Solve the Lagrange-multiplier problem for minimizing/maximizing f on

$$\text{Bd}(C) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + 4y^2 = 1\},$$

which is the boundary of C in $\mathbb{R}^2$.

(10 points)

Note: You are required to apply the Method of Lagrange Multipliers.

- (d) Explain why $f|_C$ attains an absolute minimum/maximum. **(3 points)**

- (e) Determine the absolute minimum and absolute maximum of $f|_C$. **(2 points)**

Solution**Method 1: Lagrange multipliers**

- (a) $\text{grad } f = e^{-xy}(-y, -x) = 0 \iff x = y = 0$ ($e^{-xy} > 0$), and $(0, 0) \in \text{int } C$.

The only critical point of $f|_C$ is $(0, 0)$.

- (b)

$$\begin{aligned} f_{xx} &= y^2 e^{-xy}, & f_{yy} &= x^2 e^{-xy}, & f_{xy} &= (xy - 1)e^{-xy} \\ f_{xx}(0, 0) &= 0, & f_{yy}(0, 0) &= 0, & f_{xy}(0, 0) &= -1 \\ D &= f_{xx}f_{yy} - f_{xy}^2 = -1 < 0 \end{aligned}$$

$(0, 0)$ is a saddle point of $f|_C$.

- (c) $h = x^2 + 4y^2$; $\nabla h = (2x, 8y) \neq 0$ on $\text{Bd}(C)$, which is compact. Lagrange condition $\nabla f = \lambda \nabla h$:

$$\begin{aligned} -ye^{-xy} &= 2\lambda x, & -xe^{-xy} &= 8\lambda y, & x^2 + 4y^2 &= 1 \\ x = 0 \text{ or } y = 0 &\implies x = y = 0 \text{ (contradiction)} \implies xy \neq 0, \lambda \neq 0 \\ \frac{y}{x} &= \frac{x}{4y} \implies x^2 = 4y^2 && \text{(divide)} \\ 2x^2 = 1 &\implies x = \pm \frac{1}{\sqrt{2}}, & y &= \pm \frac{1}{2\sqrt{2}} \\ x = 2y : \lambda &= -\frac{1}{4}e^{-1/4}, & x = -2y : \lambda &= \frac{1}{4}e^{1/4} \end{aligned}$$

$\left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{2\sqrt{2}}\right)$ with matched signs, and $\left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{2\sqrt{2}}\right)$.

(d) f continuous on $\mathbb{R}^2$; C closed and bounded, so compact $\implies$ (Extreme Value Theorem)

$f|_C$ attains both an absolute minimum and an absolute maximum.

(e) $f(0,0) = 1$; $xy = \frac{1}{4} \implies f = e^{-1/4}$; $xy = -\frac{1}{4} \implies f = e^{1/4}$; $e^{-1/4} < 1 < e^{1/4}$.

$$\min_C f = e^{-1/4} \text{ at } \left(\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right),$$

$$\max_C f = e^{1/4} \text{ at } \left(\frac{1}{\sqrt{2}}, -\frac{1}{2\sqrt{2}} \right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}} \right).$$

Method 2: AM–GM

On C :

$$\begin{aligned} 4|xy| &\leq x^2 + 4y^2 \leq 1 && \text{(AM–GM)} \\ \implies e^{-1/4} &\leq e^{-xy} \leq e^{1/4} \end{aligned}$$

Equality $\iff |x| = 2|y|$, $x^2 + 4y^2 = 1$: $xy = \frac{1}{4}$ at $x = 2y$ (min), $xy = -\frac{1}{4}$ at $x = -2y$ (max).

Marking scheme (25 marks)

(a) (5 marks)

- 3 marks: differentiates and solves $\text{grad } f = 0$.
- 2 marks: identifies $(0,0)$ as the sole interior critical point.

(b) (5 marks)

- 3 marks: computes the required second derivatives at $(0,0)$.
- 1 mark: obtains discriminant -1 .
- 1 mark: concludes that the point is a saddle point.

(c) (10 marks)

- 3 marks: states the Lagrange system and boundary constraint.
- 3 marks: derives $x^2 = 4y^2$.
- 3 marks: finds all four boundary points.
- 1 mark: gives the associated function values or multipliers correctly.

(d) (3 marks)

- 1 mark: states continuity of f .
- 1 mark: states compactness of C .
- 1 mark: invokes the Extreme Value Theorem.

(e) (2 marks)

- 1 mark: identifies the absolute minimum and its points.
- 1 mark: identifies the absolute maximum and its points.

Question 4**(25 Marks)****PROBLEM 4.**

(a) Evaluate

$$\int_0^1 \left[\int_0^{\sqrt{(1-x)/3}} e^{y-y^3} dy \right] dx. \quad (10 \text{ points})$$

Hint: Perform a change of order of integration.

(b) Evaluate

$$\int_0^2 \left[\int_0^{\sqrt{2y-y^2}} \sqrt{x^2+y^2} dx \right] dy. \quad (15 \text{ points})$$

Hint: Perform a transformation from Cartesian to polar coordinates.**Solution****Method 1: Fubini and polar coordinates**(a) $R = \{0 \leq x \leq 1, 0 \leq y \leq \sqrt{(1-x)/3}\} = \{0 \leq y \leq \frac{1}{\sqrt{3}}, 0 \leq x \leq 1 - 3y^2\}.$

$$\begin{aligned} \iint_R e^{y-y^3} dA &= \int_0^{1/\sqrt{3}} \int_0^{1-3y^2} e^{y-y^3} dx dy && \text{(Fubini)} \\ &= \int_0^{1/\sqrt{3}} (1-3y^2) e^{y-y^3} dy = \left[e^{y-y^3} \right]_0^{1/\sqrt{3}} && (u = y - y^3) \\ &= e^{\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{3}}} - e^0 \end{aligned}$$

$$\boxed{e^{2/(3\sqrt{3})} - 1.}$$

(b) Region: $x^2 + (y-1)^2 \leq 1, x \geq 0$. Polar: $r^2 = 2r \sin \theta, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq r \leq 2 \sin \theta$.

$$\begin{aligned} \iint \sqrt{x^2+y^2} dA &= \int_0^{\pi/2} \int_0^{2\sin\theta} r \cdot r dr d\theta && (\sqrt{x^2+y^2} = r, dA = r dr d\theta) \\ &= \frac{8}{3} \int_0^{\pi/2} \sin^3 \theta d\theta = \frac{8}{3} \cdot \frac{2}{3} \end{aligned}$$

$$\boxed{\frac{16}{9}.$$

Marking scheme (25 marks)(a) **(10 marks)**

- 3 marks: describes the region correctly after reversing the order.
- 2 marks: supplies the correct reversed bounds.
- 3 marks: integrates with respect to x and recognizes the substitution $u = y - y^3$.
- 2 marks: evaluates the endpoints correctly.

(b) **(15 marks)**

- 3 marks: recognizes and states the Cartesian disk region.

- 4 marks: obtains the correct polar bounds $0 \leq \theta \leq \pi/2$, $0 \leq r \leq 2 \sin \theta$.
- 3 marks: transforms both the integrand and area element correctly.
- 3 marks: evaluates the r -integral and $\int_0^{\pi/2} \sin^3 \theta \, d\theta$.
- 2 marks: obtains $16/9$.

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